

Seokyoung Park

Agricultural Sensing and Plant Phenotyping

✉ seokyoung.park@pusan.ac.kr | 🌐 seokyoung26.github.io

Research interests

My research interests focus on image-based analysis and sensing for agricultural applications, particularly plant phenotyping. I aim to integrate imaging, environmental sensing and robotic platforms to quantify crop traits and understand their relationships with growing conditions.

Education

Pusan National University

Mar. 2025 – Feb. 2027 (expected)

M.S. in Bio-Industrial Machinery Engineering | GPA: 4.36/4.50

Pusan National University

Mar. 2017 – Feb. 2025

B.S. in Plant Bioscience | GPA: 4.26/4.50

Minor in Bio-Industrial Machinery Engineering

Summa Cum Laude; President's List

Research experience

Precision Agriculture and Smart Sensing System Lab., Pusan National University

Jun. 2024 – Present

Roles: Graduate Researcher, Mar. 2025 – Present; Undergraduate Research Assistant, Jun. 2024 – Feb. 2025

Damage-aware robotic gripping for tomato harvesting

- Developed Python software for robotic compression experiments and force-signal processing using a myCobot 320 Pi, an adaptive gripper and a force sensor; evaluated four loading rates ranging from 6.9 to 15.3 N/s.
- Combined force-deformation measurements with hyperspectral imaging at 1 hour and 5 days after compression to assess immediate and delayed tomato damage.
- Contributed to experimental design, data analysis and manuscript preparation as first author. The resulting manuscript is under review at *Biosystems Engineering*.

Quadruped robotics for agricultural sensing

- Conducted greenhouse microclimate mapping using a Unitree Go2 quadruped robot, measuring temperature, relative humidity and CO₂ at two canopy heights across eight crop lanes.
- Produced 24 spatial maps to evaluate the representativeness of fixed greenhouse sensors and contributed to the resulting manuscript as a co-first author.

Quadruped-based 3D plant phenotyping

- As my M.S. thesis research, I am developing a mobile RGB-D workflow for close-range 3D plant reconstruction using a RealSense D456 mounted on a Unitree Go2 quadruped robot.
- Implemented depth filtering, IMU- and odometry-based pose correction, point-cloud registration, and trajectory optimization; strengthened my ability to assess reconstruction quality by evaluating local consistency using vegetation overlapping and nearest-neighbor distance.

Laboratory project: Development of an Integrated Sensing, Monitoring, and Decision-Making Model for Field Surveys in Agricultural Environments Using a Quadruped Robot (NRF/MSIT; participant).

Quadruped-based upstream reservoir surveys

- Conducted close-range upstream soil surveys using a quadruped robot, gaining field experience in robotic data collection outside controlled greenhouse settings.
- Combined RGB imaging and RTK-GNSS data to produce location-linked ground-surface datasets, developing practical skills in integrating visual and positioning data for field surveys.

Contract research: Quadruped-based Close-range Upstream Soil Survey under the Integrated Technology Development for Reservoir Embankment Risk Prediction and Maintenance Management Project, Phase I (NDMI; completed).

Air-dome foundations and structural sensing

- Uplift resistance of expandable foundation piles: Modelled wing inclination and dimension effects in Abaqus/Standard and analyzed uplift load–displacement behavior and soil plastic zones in sandy soil, strengthening my ability to interpret soil–structure interaction through finite-element analysis.
- Load-cell sensing for air-supported domes: Installed load cells and acquired cable-tension data at Damyang (15 June 2026) using an indicator, USB-to-RS485 interface, and Raspberry Pi. This work strengthened my practical skills in assembling and operating a field measurement system.

Laboratory project: Development of a Commercial-scale, Low-energy Vertical Farming System Based on Passive Air-dome Technology for Hot Climates in the Middle East (IPET/MAFRA; participant).

Functional Crop Laboratory, Pusan National University

Jan. 2023 – Mar. 2024

Role: Undergraduate Research Student

- Investigated soybean leaf phenotyping under drought using time-series RGB images, YOLOv8, and image-processing methods, with environmental measurements collected using Arduino-based sensors; developed foundational skills in image-based plant phenotyping and environmental sensing.

Publications and manuscripts

Peer-reviewed publication

[1] J. Park, **S. Park**, E. Jung, J. Song, G. Park, J. Lee, M. Kim, and J. Park. Inertial sensor applications in agriculture: Sensor types and data utilization techniques for mechanical and wearable applications. *Korean Journal of Agricultural Science*, 52(4):447–469, 2025. doi: 10.7744/kjoas.520405.

Under review

Park, S., Song, J., Hwang, H., Park, G., and Park, J. (2026). Speed-dependent control of robotic gripper for tomato harvesting to minimize compression-induced damage. *Biosystems Engineering*. **Under review.**

Manuscripts in preparation

Hwang, H.[†], **Park, S.**[†], and Park, J. Greenhouse microclimate mapping using a quadruped robot to assess fixed sensor representativeness. [†]Equal contribution.

Park, S., Park, J., Jang, J., Park, D., Kwon, S., Kim, M., and Choi, G. Analysis of uplift resistance characteristics of expandable piles in sandy soil with varying wing inclination and dimensions.

Research funding

- **Master’s Student Research Grant:** National Research Foundation of Korea (NRF), funded by the Ministry of Education (MOE) Grant RS-2025-25438074
Selected for a one-year individual research award under the NRF Research Subsidies for Master’s Students program.
Title: Development of Locomotion Algorithms and Autonomous Control Systems for Quadruped Robots Specialized for Korean Agricultural Terrain.

Technical skills

- **Programming and data analysis:** Python (NumPy, pandas and SciPy) and C/C++; time-series processing, nonlinear model fitting, statistical analysis and research visualization.
- **Robotics and mobile sensing:** ROS 2, Unitree Go2, RTK-GNSS, LiDAR and IMU; robot–sensor integration, time-synchronized data acquisition and location-linked data processing.
- **Instrumentation and data acquisition:** Force sensors, load cells, Raspberry Pi, Arduino and RS-485; experimental control, sensor calibration and CSV-based data logging.
- **Crop imaging and analysis:** RGB and hyperspectral imaging; Segment Anything Model (SAM) and YOLOv8, image segmentation, spectral comparison and time series crop-image analysis.
- **3D reconstruction for plant phenotyping:** RGB-D data acquisition using a RealSense D456; depth processing, point cloud generation and registration, pose alignment using IMU and robot odometry, and evaluation of registration consistency.
- **Finite-element modelling:** Abaqus/CAE and Abaqus/Standard; contact modeling between piles and soil, simulations under displacement control, parameter studies and interpretation of soil plastic zones.

Selected conference presentations

- **Park, S.** (14 July 2026). Height-resolved 3D microclimate mapping and spatiotemporal analysis within greenhouse crop canopies using a quadruped robot. *AKABFE Student Research Spotlight, ASABE Annual International Meeting*, Indianapolis, USA. Presenter.
- **Park, S.,** Park, J., Hwang, H., and Kang, S. (2026). Ground-reaction-force-based lateral slope estimation for furrow-following quadruped robots. *Korean Society for Agricultural Machinery Spring Conference*.
- **Park, S.,** Park, J., Song, J., and Kim, M. (31 October 2025). Analysis of pile pull-out behavior in Middle Eastern soil using digital image correlation. *Korean Society of Agricultural Engineers Autumn Conference*, Jeju.
- Hwang, H., **Park, S.,** et al. (22 May 2025). Remote control and integrated sensor-data environment for an agricultural quadruped robot. *Korean Society for Precision Agriculture Spring Conference*. Co-author.
- **Park, S.,** Park, J., Park, G., Song, J., and Hwang, H. (25 April 2025). Minimizing damage and optimizing control in fruit and vegetable harvesting using a robotic arm and gripper with real-time force sensing. *Korean Society for Agricultural Machinery Spring Conference*.

Honours and awards

- **Best Paper Presentation Award (team)**, Korean Society for Precision Agriculture, May 2025 and November 2024.
- **Excellent Paper Presentation Award**, Korean Society for Agricultural Machinery, April 2025 and October 2024.
- **Excellence Award**, PNU–PKNU Field Problem Research Group Final Symposium, November 2024.
- **Best Presentation Award (team)**, Korean Society for Engineering Education, September 2024.
- **Bronze Prize**, PNU College of Natural Resources and Life Science Capstone Design Competition, October 2023.
- **Excellence Award**, Busan LINC 3.0 Entrepreneurship Shared University Competition, January 2023.
- **PNU merit scholarships**: four full-tuition and two partial-tuition awards, 2017–2024.

Selected coursework

Graduate: Biological Image Processing; Statistical Processing and Data Analysis in Bio-Industrial Machinery Engineering; Advanced Bio-Industrial Machinery Design; Biosystems Mechanics; Advanced Biomaterials Engineering I.

Undergraduate – engineering and sensing: Biosensor and Bio-Instrumentation & Practice; Biosystems Control Engineering; Bio-Production Mechatronics Design; Applied Mechanics of Materials; Bio-Industrial Machine Element Design (I); Food Process Engineering & Practice.

Undergraduate – programming and quantitative methods: Basic Computer Programming; Experimental Design for Biologists; Biostatistics; Calculus (II).

Undergraduate – plant science and cultivation: Plant Physiology and Practice; Plant Genetics; Plant Molecular Biology; Plant Biotechnology; Plant Breeding (I, II); Soil Science; Plant Ecology; Protected Horticulture.

Additional training

Modeling Contact with Abaqus/Standard, VPK PLM Solution Training Center, 25–26 June 2026.

Introduction to Abaqus/CAE, VPK PLM Solution Training Center, 18–20 March 2026.